

Kyle Webb

Staff Data Scientist · 10+ years · statistics, data engineering, and applied AI

CONTACT

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PROFESSIONAL SUMMARY

Data scientist and data engineer, 10+ years across research, federal health, and industry. Builds both sides: trustworthy data platforms and governance, and the ML and statistics that turn data into decisions.

SELECTED PROJECTS

30-day hospital readmission risk

Production-style ML system, FHIR-native pipeline, deployed model, drift monitoring

SS beneficiary fraud

Bayesian estimate from public SSA records

TECHNICAL SKILLS

Python · R · SQL · Linux · Git · Terraform · AWS · Databricks · Spark/PySpark · Airflow (MWAA) · dbt · Docker · MLflow · scikit-learn · LLMs

CERTIFICATIONS

Databricks Data Engineering Associate (2026)
AWS Solutions Architect – Associate (2026)
AWS Cloud Practitioner (2023)

EDUCATION

M.S., Statistics

Virginia Tech (2015–2016)

B.S., Statistics

Wake Forest (2011–2015)
GPA 3.4/4.0

IMPACT HIGHLIGHTS

- Slashed median EMR workload runtime **82%** and cost **93%** by migrating legacy workloads to Databricks with automated validation, testing, and monitoring.
- Migrated **approximately 191K tables** from AWS Glue to Unity Catalog and transitioned SAS-centric pipelines to modular Python/Spark, cutting compute unit costs **50%**.
- Delivered validated data distributions across **20 CDR sources** (terabytes of claims, provider, and beneficiary data) with a six-engineer DevOps/data team.
- Architected data governance with strict role-based access for **2,000+ users** across a Kinesis-driven repository, and launched Delta Shares to end duplicate data copies.
- Cut governance group onboarding from a **48-hour** SLA to under **15 minutes** for standard access and workspace readiness.
- Built an applied-AI showcase: Databricks Apps cost-analysis dash with live natural-language Q&A (**Claude Sonnet 4.6**) via MCP server, pytest + Playwright tested.

EXPERIENCE

For roles before 2018, see the [full CV](#).

eSimplicity — Staff Data Scientist

Sept 2023 – Present · Remote

- Developed a copilot agent that standardizes data-pipeline deployments on Databricks.
- Co-led FinOps and governance frameworks managing approximately \$167K in monthly spend.
- Modernized legacy metadata formats into a data catalog application built on a customized DataHub fork.
- Ran 20+ technical interviews for data science and engineering roles, assessing coding proficiency and system design.

eSimplicity — AI Data Analyst Fellow

July 2026 – Present · Remote

Rapid Innovation Center · internal applied-AI fellowship

- Managed a document reader agent that authenticates with HARP OKTA credentials to ingest Confluence, GitHub, and document stores into the Katalyst taxonomy (internal knowledge graph).
- Established program taxonomy using the eS Katalyst system, creating declarative knowledge graphs expressed as YAML files in git.
- Presented technical marketing research on AI SLA refinement for a CMS bid.

eSimplicity — Senior Data Scientist

Apr 2022 – Aug 2023 · Remote

- Created Bash/Python tooling migrating data and metadata from Hadoop to S3 and Hive Metastore to AWS Glue, including parallelized cross-account object migration on EMR.
- Presented applied analytics and data-science workflows in R on Databricks to 150+ CDR users at CCSQ 2023 DataCamp.
- Ran a lineage-tracking pilot with OpenLineage, dbt, and Marquez to evaluate metadata capture and visualization.

GovernmentCIO — Data Scientist

2021 – 2022 · Remote

- Automated VA IT infrastructure reporting in R/Python as the team's sole developer; briefed major incidents to 200+ attendees including the VA CIO.
- Delivered analysis for 30+ briefs for the largest U.S. healthcare network, several on one-day turnarounds.

NHGRI — Bioinformatics Scientist

2020 – 2021 · Bethesda, MD

- Stood up the lab's N3C data-enclave workspace from scratch; secured data use agreements for COVID-19 EHR pipelines.
- Curated EHR data from millions of patients for a COVID-19 Phenotype-Wide Association Study (PheWAS).

NIAID — Data Science Fellow

2018 – 2020 · Bethesda, MD

- Built ETL pipelines over 4TB+ of Cerner data and parallelized HPC jobs training ML classifiers for rare diseases.
- Extended the open-source R package *ihazr* with semiparametric hazard regression for survival analysis.